The Cheon–Hwang Sub-Dittert Conjecture at $k = 3$ and $k = 4$

with a stability form of the Tverberg–Friedland theorem
and a decomposition of the deficit uniform in n and k

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Abstract. Cheon and Hwang conjectured in 1992 that $E_k(r) + E_k(c) - P_k(A) \le 2 - k!/n^k$ for every non-negative $n \times n$ matrix of total sum n and every $1 \le k \le n$, with equality only at J_n/n; the endpoint $k = n$ is Dittert's conjecture, and the intermediate range $2 < k < n$ appears to have gone untouched. One structural fact organises what follows: the deficit has degree k, not n. Where k is small enough that a fixed-degree Positivstellensatz ansatz still matches it, the symmetry-reduced programme has the *same* 12×19 shape at every dimension and is solved once over $\mathbb{Q}(n)$; where it is not, only inequalities uniform in n reach, and a local method takes over.

On the first route we settle the line $k = 3$ for every $n \ge 4$ — the inequality, the equality case, and a quantitative strengthening $(2 - 6/n^3) - \Phi_3(A) \ge \theta_2(n)\|A - J_n/n\|_F^2$ with θ_2 an explicit rational function — by a single certificate whose nineteen coefficients are rational functions of n, its two $n^2 \times n^2$ Gram matrices reduced in closed form to ten rational functions decided by Sturm sequences. This is machine-checked end to end (`subDittert_k3_full`).

The second route is local: confinement traps any violator in a collar of the Birkhoff polytope, a collar matrix splits orthogonally into a line-sum block and a doubly centred block, the centred block is governed by a *stability form of the Tverberg–Friedland theorem* proved here, and the five cross terms between the blocks have exact reductions. That yields the Cheon–Hwang inequality at $k = 4$ for every $n \ge 10$ with its equality case. The stability theorem — that σ_k on the doubly stochastic polytope exceeds its minimum by at least $\binom{n}{k}^2 c(n,k)\|A - J_n/n\|_F^2$, with c optimal to within a factor two — is kernel-checked at $k = 2$, $k = 3$ and $k = 4$ over the whole range it states for each, and is of independent interest. Fixed-dimension certificates settle $(k = 4, n = 5, 6, 7)$, so on the line $k = 4$ exactly two cells remain open, $n = 8$ and $n = 9$. Every result is stated with the layer it rests on — kernel-checked in Lean 4 at a pinned commit, exact rational verifier, or verified computation — and the grading is part of the claim.

1 The conjecture, and the shape of the problem

Let $K_n = \left\{ A \in \mathbb{R}^{n\times n} : A_{ij} \ge 0, \sum_{i,j} A_{ij} = n \right\}$, and for $A \in K_n$ let r and c be its vectors of row and column sums. Write $\sigma_k(A)$ for the sum of the permanents of all $k \times k$ submatrices of A — rows chosen from one k-subset, columns from another, independently — and put

$$E_k(v) = \frac{e_k(v)}{\binom{n}{k}}, \quad P_k(A) = \frac{\sigma_k(A)}{\binom{n}{k}^2}, \quad \gamma(n,k) = \frac{k!}{n^k}, \quad \Phi_k(A) = E_k(r) + E_k(c) - P_k(A), \tag{1}$$

with e_k the elementary symmetric function.

The conjecture (Cheon and Hwang [1], 1992). For every $A \in K_n$ and every $1 \le k \le n$, $\Phi_k(A) \le 2 - \gamma(n,k)$, with equality only at $A = J_n/n$.

At $k = n$ one has $\sigma_n = \mathrm{per}$ and $E_n(v) = \prod_i v_i$, so the statement is $\prod_i r_i + \prod_j c_j - \mathrm{per}(A) \le 2 - n!/n^n$: *Dittert's conjecture* verbatim, Conjecture 28 of the Cheon–Wanless survey [5]. At $k = 1$ the inequality is an identity *on K_n* — not on all of $\mathbb{R}^{n\times n}$ — since the difference is a constant multiple of $\sum_{ij} A_{ij} - n$; a test that treats it as a formal identity fails. The cases $k \le 2$ for every n, and every k at $n \le 3$, are known. The interesting range is $2 < k < n$, and it is that range which appears to have gone untouched (§1.4).

1.1 One decomposition, exact and uniform in n and k

Everything below is read off a single identity.

Theorem C (the deficit, decomposed uniformly in n and k). Write $A = J_n/n + b$, let R and C be the row and column sums of b, and put

$$s_d = \frac{[k]_d}{[n]_d}, \quad t_d = s_d^2 \cdot \frac{(k-d)!}{n^{k-d}} \tag{2}$$

with $[x]_d$ the falling factorial. Then for every $1 \le k \le n$, identically in b,

$$(2 - \gamma(n,k)) - \Phi_k(A) = \sum_{d=1}^{k} [t_d \cdot \sigma_d(b) - s_d \cdot (e_d(R) + e_d(C))]. \tag{3}$$

Kernel-checked: `SubDittertUniversal.universal_identity`.

(3) is elementary and no novelty is claimed for it: it is the σ_k analogue of $\mathrm{per}(A + xJ_n) = \sum_j x^{n-j}(n-j)!\,\sigma_j(A)$ combined with $e_k(\mathbf{1} + R) = \sum_d \binom{n-d}{k-d} e_d(R)$. Its role is to make every k-dependence explicit and finite, so that facts looking separate at separate k become one identity read at one degree; the absence of a $d = 0$ term is why the constant $2 - \gamma(n,k)$ is exactly right at every n and k at once. One reading is used throughout: the coefficient of a degree-d monomial takes one of three values — $-2s_d + t_d$ if the cells form a partial permutation, $-s_d$ if they have distinct rows or distinct columns but not both, and 0 otherwise.

1.2 The organising fact, and the map of the paper

Write $F(b) = (2 - \gamma(n,k)) - \Phi_k(A)$ in the centred coordinates $b = A - J_n/n$. Then $\deg F = k$, *not* n, and that is the fact deciding which method reaches where. A Positivstellensatz ansatz with a Gram basis of degree e produces terms of degree $2e + 1$, so it matches F exactly when $e = \lceil (k-1)/2 \rceil$, and the symmetry-reduced programme then has a shape depending on k alone. At $k = 3$ that shape is 12 equations in 19 unknowns at *every* dimension. At $k = 4$ the basis degree steps to $e = 2$, the programme grows to 440 unknowns, and no family in n is known for it.

region	method, and why it is the one that reaches	what it yields
$k = 3$, every $n \ge 4$	one certificate over $\mathbb{Q}(n)$: at $e = 1$ the reduced programme is 12×19 at every dimension, so a single symbolic solve covers all n	Theorem A — inequality, equality case *and* a stability bound; kernel-checked
$k = 4$, every $n \ge 10$	the local method: confine, split, apply stability on the slice, assemble on the collar. At $k = 4$ no certificate family in n exists, and only n-uniform inequalities reach	Theorem H — inequality and equality case; exact-verifier grade
$k = 4$, $n = 5, 6, 7$	exact per-cell certificates: at small n an exact solve finishes, and it gives the sharpest statement available per cell	anchor grade; the cells $(k = 4, n = 8)$ and $(k = 4, n = 9)$ remain open

Table 1: Which method is used where, and why. The stability theorem (Theorem G) is the shared engine of the middle row.

The three rows are not three papers. Both methods run on (3) — the certificate is a Positivstellensatz for the whole of F, the local method bounds (3) layer by layer — and Theorem G, the middle row's engine, is a statement about the same σ_k on the doubly stochastic slice.

1.3 Results

Theorem A ($k = 3$, every $n \ge 4$). For every $n \ge 4$ and every $A \in K_n$,

$$E_3(r) + E_3(c) - P_3(A) \le 2 - \frac{6}{n^3}, \tag{4}$$

with equality if and only if $A = J_n/n$. More precisely, with $\|\cdot\|_F$ the Frobenius norm,

$$\left(2 - \frac{6}{n^3}\right) - [E_3(r) + E_3(c) - P_3(A)] \ge \theta_2(n) \cdot \|A - J_n/n\|_F^2, \quad \theta_2(n) = \frac{n^4 + 40n^2 - 84n + 40}{n^5(n-1)^3(n-2)}, \tag{5}$$

and $\theta_2(n) > 0$ for every $n \ge 4$.
Kernel-checked: `SubDittertK3.subDittert_k3_full`, all three parts in one theorem.

The three statements are one: (5) implies both the others, since $\theta_2(n) > 0$ and $\|A - J_n/n\|_F$ vanishes only at J_n/n, while $\Phi_3(J_n/n) = 2 - 6/n^3$ exactly. We state all three because the quantitative form is what the certificate produces, and it is stronger than the conjecture asks: the functional falls away from its maximum at least quadratically in the distance to the maximiser, at an explicit rate not claimed optimal (§5).

Corollary B (two layers, stated as such). With Hwang's theorem for Dittert at $n = 3$ [3], the case $k = 3$ holds with its equality case for every $n \ge 3$, and since $k = 3$ requires $n \ge 3$ no dimension is left open on that line.

Not a single-layer result; do not cite it as one. The range $n \geq 4$ is Theorem A and is kernel-checked; the case $n = 3$ is Hwang's refereed theorem of 1987 [3], which we do not reprove. The corollary's support is Lean *plus* the published literature.

Theorem D ($k = 2$, every $n \geq 2$). $\Phi_2(A) \leq 2 - 2/n^2$ for every $A \in K_n$. With $\kappa = 2/(n(n-1))$ the deficit is a manifest sum of squares on the hyperplane $\sum_{ij} b_{ij} = 0$:

$$(2 - 2/n^2) - \Phi_2(A) = \frac{1}{2}\left[\kappa^2\|b\|^2 + \kappa(1-\kappa)(\|R\|^2 + \|C\|^2)\right]. \tag{6}$$

Kernel-checked: `SubDittertK2.subDittert_k2`. *The statement is classical* (§1.4); the contributions are the derivation from (3) and the machine-checked proof.

Theorem D needs no Cauchy–Schwarz, no Gram matrix and no Maclaurin inequality: (6) falls out of (3) at $k = 2$, both coefficients being non-negative for $n \geq 2$ because $\kappa \leq 1$ there. Its interest is as a control on the uniform machinery — the one case where that machinery runs end to end against an independently known answer. The same route does not reach $k = 3$, where the deficit is not a sum of squares on the hyperplane.

The next two results are recalled rather than claimed; it is the formalisation, not the mathematics, that is ours.

Theorem E (confinement; Cheon and Wanless [6], Theorem 2.1, at $k = n$; transferred to all k). Let $2 \leq k \leq n$ and $A \in K_n$. Then

$$\frac{\|r - \mathbf{1}\|^2 + \|c - \mathbf{1}\|^2}{n(n-1)} - \gamma(n,k) \leq (2 - \gamma(n,k)) - \Phi_k(A), \tag{7}$$

so any A violating the Cheon–Hwang bound satisfies $\|r - \mathbf{1}\|^2 + \|c - \mathbf{1}\|^2 \leq (n-1)k!/n^{k-1}$.
Kernel-checked: `SubDittertMaclaurin.theoremM'`, `confinement'`.

Theorem F (Newton; Maclaurin). For every real-rooted real polynomial and every admissible index Newton's inequality holds; in vector form, for $v \in \mathbb{R}^n$ and $1 \leq j \leq n-1$, $e_{j-1}(v)\, e_{j+1}(v)\binom{n}{j}^2 \leq e_j(v)^2\binom{n}{j-1}\binom{n}{j+1}$, with no positivity hypothesis. Consequently, for $v \geq 0$ with $\sum_i v_i = n$ and $2 \leq k \leq n$, $E_k(v) \leq E_2(v)$.
Kernel-checked: `NewtonIneq.newtonAt_all`, `newton_esymF`, `pnorm_le_two`. *Classical mathematics*; the contribution is the formalisation. Neither inequality is in Mathlib v4.14.0, whose `NewtonIdentities` carries Newton's *identities* only.

The remaining two results are the engine and the conclusion of the local route. The first lives on the doubly stochastic polytope Ω_n and is a quantitative form of a theorem conjectured by Tverberg [17] and proved by Friedland [18]: σ_k on Ω_n attains its minimum only at the barycentre, the case $k = n$ being the van der Waerden conjecture settled by Egorychev [20] and Falikman [21]. Those theorems give strictness but no rate.

Theorem G (stability of the Tverberg–Friedland minimum). Let $2 \leq k \leq 5$ and let n satisfy

$$n \geq 2 \text{ at } k = 2, \quad n \geq 4 \text{ at } k = 3, \quad n \geq 8 \text{ at } k = 4, \quad n \geq 14 \text{ at } k = 5. \tag{8}$$

Then for every doubly stochastic $n \times n$ matrix A,

$$\sigma_k(A) - \binom{n}{k}^2 \frac{k!}{n^k} \geq \binom{n}{k}^2 c(n,k)\, \|A - J_n/n\|_F^2, \quad c(n,k) = \frac{k(k-1)\,k!}{4n^k(n-1)^2}. \tag{9}$$

The constant is optimal to within a factor two (Proposition S5), and the cell $(k, n) = (3, 3)$ is a genuine exception with an explicit witness.
Kernel-checked at ($k = 2$, every $n \geq 2$) and ($k = 3$, every $n \geq 4$) — `stabilityAt_two`, `stabilityAt_three` in `StabilityK3.lean` at commit `1507013` — and at ($k = 4$, every $n \geq 8$) — `stabilityAt_four` in `StabilityK4.lean` at commit `944b517` — in each case over the whole range the theorem states for that k. *Written proofs plus the exact verifier* `graded_verify_stability.py` at ($k = 5, n \geq 14$): no `stabilityAt_five` exists. §5 states the division and its two limits.

Theorem H ($k = 4$, every $n \geq 10$). For $k = 4$ and every $n \geq 10$, $\Phi_4(A) \leq 2 - 24/n^4$ for every $A \in K_n$, with equality only at $A = J_n/n$.
Not formalised. Support: the written proofs of §3.3 plus the exact verifier `graded_verify_k4.py`, which recomputes every displayed quantity of that part over $\mathbb{Q}$ with no floating-point arithmetic in any decision.

Between Theorem A's line and Theorem H's range sit the cells ($k = 4, 5 \leq n \leq 9$). Three are settled by fixed-dimension certificates at anchor grade — ($k = 4, n = 5$), ($k = 4, n = 6$) and ($k = 4, n = 7$) — so the open cells on the line $k = 4$ are exactly $n = 8$ and $n = 9$. §4 records all five at the grade each has reached and states that gap once.

1.4 Support layers, and prior work

Every result above is stated with the layer it rests on, because in this corner of the literature the distinction has recently been expensive. Theorems A and C–F are proved in Lean 4; Theorem G splits by cell as its statement records; Theorem H is not formalised; the fixed-dimension certificates of §4 are exact rational data accepted by an independent standalone verifier, and no theorem here depends on them. Nothing here is refereed, including this paper — exact verification and machine checking are different things from refereeing, and weaker ones. §5 is the precise record.

The intermediate range has attracted almost nothing. The Cheon–Hwang paper [1] has five citing papers — [5], [6], [7] and the two of [12] — none of which works on the generalisation, and to our knowledge neither the literature nor any public code repository carries a result on the intermediate range: the one adjacent package, inside a repository of [11], explicitly disclaims the unresolved intermediate cases, and that disclaimer is also the clearest available statement of the prior status of Theorem D. The endpoint $k = n$ is not attacked here and its status is not this paper's subject — one public unrefereed repository of [11] claims Dittert in full, on which we take no view — and every priority claim made here is narrowed to what the Lean development and the exact verifiers actually carry. Reference [7] is paywalled and was not obtained; it is reported to settle $n \le 3$ for all k, and nothing here depends on it.

2 The certificate route: $k = 3$, uniformly in n

2.1 The objective, and the ansatz

A certificate for the wrong polynomial is worthless and looks identical to a certificate for the right one, so F is checked before anything else: σ_k by two structurally different algorithms (enumeration of the $\binom{n}{k}^2$ index pairs, against $\operatorname{per}(A + xJ_n) = \sum_j x^{n-j}(n-j)!\sigma_j(A)$ in $\mathbb{Q}[x]$ with the order-n permanent by Ryser), and the polynomial built at $(n,k) = (4,4)$ compared coefficient by coefficient — 1040 monomials, zero differences — against an independent pipeline for the $k = n$ case that already underlies a verified certificate. One trap deserves naming, since the obvious check falls into it: $2 - \gamma(4,3) = 2 - \gamma(4,4) = 61/32$ and $2 - \gamma(5,4) = 2 - \gamma(5,5) = 1226/625$, so seeing the right *bound* is no evidence at all that the $k = 3$ code does what it claims. Whole polynomials separate the cases; at $n = 4$ the two objectives disagree at every one of their 1040 monomials.

In centred coordinates the ansatz is

$$F(b) = \sigma_0(b) + \sum_p \left(\frac{1}{n} + b_p\right)\sigma_p(b) + \lambda(b)\cdot\left(\sum_q b_q\right), \tag{10}$$

where σ_0 and every σ_p is a sum of squares and λ is free. On K_n the last term vanishes and every $1/n + b_p = A_p \ge 0$, so the right side is a sum of non-negative terms and $F \ge 0$ there, which is the theorem. The shape is Putinar's [2], truncated at a fixed degree, but no existence theorem is invoked and none applies: Putinar's needs strict positivity, and F vanishes at J_n/n in the relative interior of K_n — the representation is exhibited, not deduced. Sum-of-squares relaxations are Lasserre's hierarchy [13] and block-diagonalising by a symmetry group is Gatermann–Parrilo [14], both used in their plain forms; what is not taken from that line is the coefficient field, since the programme is posed and solved over $\mathbb{Q}(n)$, one symbolic solve whose output is a certificate for every n at once. Optimisation uniform in the dimension is itself an active area [15].

Why the degree closes. At $k = 3$ a Gram basis of degree 1 gives $\deg\sigma \le 2$ and $\deg(\sigma_p b_p) \le 3 = \deg F$, with no surplus top band to cancel; the Gram matrices are $n^2 \times n^2$. *Why rounding works.* The standing obstruction to exact rational rounding is that a tight bound forces a singular optimal Gram. Here the Hessian of F at J_n/n restricted to $\{\sum X = 0\}$ is positive definite — at $(n,k) = (4,3)$ its characteristic polynomial there is $x(x - 1/16)^9(x - 29/16)^6$ over $\mathbb{Q}$, the multiplicities $(n-1)^2$ and $2(n-1)$ forced by Schur's lemma — so centring at the extremiser and excluding the constant monomial removes the only degenerate direction.

2.2 The programme's shape is fixed in n

The problem is invariant under $(S_n \times S_n) : \mathbb{Z}_2$ acting by row permutations, column permutations and transposition. Reducing (10) by that group leaves, at every n alike, 12 orbit constraint rows of rank 11 over $\mathbb{Q}(n)$, and $3 + 11 + 5 = 19$ unknowns (σ_0, σ_{11} and λ); only the cone size n^2 grows. Nineteen unknowns and twelve equations, at $n = 4$, $n = 5$, $n = 6$ and beyond. This is representation stability, and here — unlike at $k = n$, where the same stability is undercut by the growing degree — nothing counteracts it.

Both halves of the 12×19 system are *derived* as functions of n, not interpolated; "no fit is used anywhere" is a stronger statement than "the fit was validated". The coefficient in F of an arbitrary monomial is the $k = 3$ case of the three-value rule of (3), and the orbit sizes come from orbit–stabiliser, every one a polynomial in n of degree at most 6 because a multiset of at most three cells meets at most three rows and three columns. Both were checked against the fully expanded objective at $n = 4, 5, 6$ over the *whole* coefficient vector, absent monomials included — 969, 3276 and 9139 monomials, no mismatches — and the assembled system was compared entry by entry with the one built by the original code path, which shares no logic with it and which produced the already-verified fixed-n certificates. Row-reducing over $\mathbb{Q}(n)$ gives rank 11, one dependent row, *consistent*, with an 8-dimensional affine family of solutions: the linear half of the certificate exists at every n at once.

2.3 Definiteness, and the design problem

What remains is to choose within that family so that both Gram matrices are positive definite for every $n \ge 4$. Both are block-diagonalised *in closed form* — a numerical basis at one n could not settle all n. The σ_0 Gram lies in the Bose–Mesner algebra of the rook's graph $K_n \square K_n$, so it has exactly three eigenvalues $\theta_0, \theta_1, \theta_2$, of multiplicities 1, $2(n-1)$, $(n-1)^2$. For σ_{11}, with V' the standard $(n-2)$-dimensional representation of S_{n-1}, $\mathbb{R}^{n^2} = 4(1|1) \oplus 2(V'|1) \oplus 2(1|V') \oplus (V'|V')$; transposition fuses $(V'|1)$ with $(1|V')$ and splits the trivial multiplicity space as $3+1$, leaving a 3×3 block A, a sign block B, a 2×2 block C of multiplicity $2(n-2)$ and a 1×1 block D of multiplicity $(n-2)^2$. Both consistency checks pass: $6+1+3+1 = 11$ orbit parameters, and $3+1+4(n-2)+(n-2)^2 = n^2$. The decomposition was verified against a direct eigendecomposition of the assembled matrix on *random* orbit coefficients at $n = 4, ..., 8$; a check run on the real certificate could pass by accident on a matrix carrying extra structure.

> *Consequence.* Positive definiteness of both Grams for all $n \geq 4$ is exactly the positivity of *ten* explicit rational functions of n: the three σ_0 eigenvalues, the three leading principal minors of A, the block B, the two leading minors of C, and the block D. One is not independent: $D = \theta_2/n$ *exactly*, an identity of rational functions and not a coincidence at sampled n, so the certificate rests on *nine* independent positivity facts. We certify all ten regardless, since Sylvester's criterion asks for all ten directly.

Choosing the eight free variables as functions of n is the whole difficulty, and its shape is not the obvious one. The feasible set is *unbounded*; worse, the quantities that must be controlled need the two Grams positive semidefinite only on $\mathbf{1}^\perp$, which leaves a four-dimensional *lineality space* — exactly the λ reparametrisation — along which optima drift so violently that no curve in n can follow them. Quotienting it out by a transversal slice makes the set compact and leaves an essential design problem in four unknowns, and that is necessary but not sufficient, because *the compact set that remains is a sliver*: in scaled coordinates $\beta = f \cdot n^3$, four of the ten quantities are differences of terms of size n^2 or n^3 that must cancel to $O(1)$, which forces $\beta_6 - 2\beta_9 = O(n^{-2})$, $\beta_{12} - (2\beta_9 - 1) = O(n^{-1})$ and $\beta_{11} - 2\beta_{12} - 2 = O(n^{-1})$, the last pinned to $O(n^{-3})$. A least-squares fit whose residual is small against the diameter of that set is still far outside it.

The fix is to write the cancellations into the coordinates so that they happen symbolically rather than numerically — $\beta_9 = b$, $\beta_6 = 2b + x/n^2$, $\beta_{12} = 2b - 1 + y/n$, $\beta_{11} = 2\beta_{12} + 2 + z/n$ — and then *solve for z exactly over $\mathbb{Q}(n)$ from the equation $\theta_2 = D$* rather than fitting it. That is the step which makes the difference: the two quantities are affine in z with opposite-sign coefficients of size n^2 and the window between them has width $O(n^{-2})$, so at $n = 10^6$ a numerical fit would need twelve correct digits merely to stay inside. With z eliminated the remaining parameters carry $\Theta(1)$ coefficients and an ordinary semidefinite program over a grid of n finds them:

$$b = 1, \quad x = 8 + \frac{20}{n}, \quad y = -2 - \frac{10}{n}, \quad z = \frac{6n^7 - 28n^6 + 41n^5 - 28n^4 + 48n^3 - 164n^2 + 208n - 80}{n^7 - 7n^6 + 19n^5 - 25n^4 + 16n^3 - 4n^2}. \tag{11}$$

The transferable form of the step is short: *in a family of certificates indexed by a parameter, identify the tight direction and eliminate it symbolically; fit only the slack ones.* The nineteen resulting certificate variables are exact rational functions of n, recorded in the accompanying material; the largest has numerator of degree 9 over denominator of degree 12, and θ_2 decays like n^{-5}, so the endpoint costs margin, not sign. Four routes a reader would otherwise try are closed by exact rational witnesses, recorded with those witnesses in the kit: linear programming for the design step, which is bilinear rather than linear once one sees that the rescaling it needs must be read off the certificate it is trying to choose; least squares through analytic centres, which fails off-grid at $n = 17$, 71 and 811; pinning four entries to constant targets, which fails for all 210 four-subsets; and restricting λ to a sum of squares, which removes the recession cone but not the lineality space.

2.4 Positivity decided, and verified

Substituting $n = m + 4$ turns each of the ten quantities into a ratio of polynomials in m on the range $m \geq 0$. The sign of each denominator there is *checked*, not assumed, and the numerator's positivity on $[0, \infty)$ is then *decided* by a Sturm chain on the squarefree part, comparing sign variations at 0 and at $+\infty$. There is no sufficiency gap in that step, which is why the tempting shortcut "all coefficients non-negative in m" is only a heuristic: $m^2 - m + 1$ is positive everywhere and has a negative coefficient.

	`theta0`	`theta1`	`theta2`	`A_1`	`A_2`	`A_3`	`B`	`C_1`	`C_2`	`D`
sqfree deg	0	6	4	0	5	13	5	5	12	4
$V(0) = V(\infty)$	—	1	1	—	1	3	2	1	2	1
roots in $(0, \infty)$	0	0	0	0	0	0	0	0	0	0

Table 2: All ten quantities, positive for every $n \geq 4$. Each column is a complete decision, not a sample. `D` and `theta2` share a numerator — visible in the matching squarefree degree — because $D = \theta_2/n$ exactly. All ten denominators are sign-definite on $n \geq 4$, as are those of all nineteen certificate variables, so nothing blows up at any dimension.

A certificate is worth what its checking is worth. The independent verifier *re-derives the objective from the 1992 definition*, using none of the closed forms, none of the block decomposition and none of the Sturm machinery, and checks at each n that both Grams are *positive definite* by exact rational LDL^T on the *full* $n^2 \times n^2$ matrices rather than through the blocks — so it confirms the block theory of §2.3 as well as the numbers — that (10) holds at random rational points, and that the constant is $2 - k!/n^k$. It accepts at $n = 4, 5, 6, 7$ and $n = 12$ exactly over $\mathbb{Q}$, and at $n = 25$ with the identity in $\mathbb{F}_p$ at three primes near 2^{20}. It also accepts the independently produced stored certificates at $n = 4, 5$ as a positive control, and rejects four single-variable perturbations; a verifier that never rejects proves nothing.

3 The local method: confinement, stability, and the collar

Where the certificate route stops, the following chain takes over. It has three links and one assembly: confinement traps a violator near the Birkhoff polytope; on the polytope itself the centred core is controlled by Theorem G; and the collar between them is paid for by exact reductions of the cross terms.

3.1 Confinement, for every k

Theorem E is recalled from [6] and transferred to all k; nothing here is claimed as new mathematics. It is included because it is uniform in k where the certificate is not, and short enough to formalise once for the whole family. The proof needs no expansion of the objective. Split the deficit as $(2 - \gamma(n,k)) - \Phi_k(A) = [1 - E_k(r)] + [1 - E_k(c)] + [P_k(A) - \gamma(n,k)]$, where the third bracket is at least $-\gamma(n,k)$ because P_k of a non-negative matrix is non-negative. For the first two, Maclaurin in the telescoped form of Theorem F gives $E_k(r) \le E_2(r)$ on the simplex, and E_2 is exactly computable, $E_2(r) = 1 - \sum_i (r_i - 1)^2/(n(n-1))$ whenever $\sum_i r_i = n$ (`SubDittertK2.E_two_eq`, needing no positivity). The contrapositive is the confinement statement: a violator has its line sums within $(n-1)k!/n^{k-1}$ of $\mathbf{1}$ in squared ℓ^2 distance, a neighbourhood of the *Birkhoff polytope* — not of J_n/n; the doubly centred directions are not confined — that shrinks in n for every fixed $k \ge 2$.

The telescoped form $E_k \le E_2$ carries no fractional powers, unlike the usual $E_k^{1/k} \le E_2^{1/2}$, and that is what makes it the statement to formalise; its Lean proof, and a second independent one by a different route kept deliberately outside the build, are described in §5.

3.2 Stability on the doubly stochastic polytope

Everything in this subsection is confined to Ω_n: the estimates use four a-priori facts that hold there and fail on the larger set K_n, and nothing here should be read as a statement about K_n. The transfer to the collar, and the degradation it costs, is §3.3.

Fix $n \ge 2$, let $A \in \Omega_n$ and put $B = A - J_n/n$, $Q = \|B\|_F^2$. Every row and column of B sums to zero; we call this the *centred slice*. Write $q_i = \sum_j b_{ij}^2$, $q'_j = \sum_i b_{ij}^2$, $M = \max(\max_i q_i, \max_j q'_j)$, $\beta = 1 - 1/n$, $p_r = \sum_{i,j} b_{ij}^r$, and $Y_R = \sum_i q_i^2$, $Y_C = \sum_j q_j'^2$, $Z = \left\|B^T B\right\|_F^2$. Four invariants appear at degree five: with $f_3(i) = \sum_j b_{ij}^3$, $g_3(j) = \sum_i b_{ij}^3$ and q, q' read as vectors, $\Gamma_a = q^T B q'$, $\Gamma_b = \sum_{i,j} b_{ij}^2 (BB^T B)_{ij}$, $\Gamma_c = \sum_i q_i f_3(i)$ and $\Gamma'_c = \sum_j q'_j g_3(j)$. Finally s_m, t_m are the coefficients of Theorem C in the index letter m of the layer they weight; two facts about them are used repeatedly,

$$t_2 = \frac{k(k-1)\,k!}{n^k(n-1)^2} \quad \text{so} \quad c(n,k) = t_2/4, \quad \text{and} \quad \frac{t_{m+1}}{t_m} = \frac{n(k-m)}{(n-m)^2}. \tag{12}$$

> **Lemma S1 (the layer identity).** For every $A \in \Omega_n$ and every $1 \le k \le n$, with $B = A - J_n/n$,
>
> $$\frac{\sigma_k(A)}{\binom{n}{k}^2} - \frac{k!}{n^k} = \sum_{m=2}^{k} t_m\, \sigma_m(B). \tag{13}$$
>
> Kernel-checked at *every* k: `layer_identity` in `LayerIdentity.lean` (commit `27bda3f`), under hypotheses weaker than this part needs — see §5.

Lemma S1 is the P_k half of (3) read on the centred slice, where the e_d half vanishes. *Proof.* Expanding a permanent of a sum, for index sets α, β of size k, $\operatorname{per}((X+Y)[\alpha|\beta]) = \sum \operatorname{per}(X[S|T]) \operatorname{per}(Y[\alpha \setminus S \mid \beta \setminus T])$ over $S \subseteq \alpha$, $T \subseteq \beta$ with $|S| = |T|$. Take $X = J_n/n$, $Y = B$, so $\operatorname{per}(X[S|T]) = d!/n^d$ for $|S| = |T| = d$. Summing over all α, β and writing $m = k - d$, each pair (U, V) with $|U| = |V| = m$ is completed to (α, β) in $\binom{n-m}{k-m}^2$ ways, so $\sigma_k(A) = \sum_m \binom{n-m}{k-m}^2 \big((k-m)!/n^{k-m}\big)\sigma_m(B)$; dividing by $\binom{n}{k}^2$ turns the coefficient into t_m. The $m = 0$ term is $t_0 = k!/n^k$ and the $m = 1$ term vanishes because $\sigma_1(B) = \sum_{i,j} b_{ij} = 0$. ■

Since $t_m > 0$ and $\sigma_2(B) = Q/2$ on the slice, Theorem G is equivalent to

$$\sum_{m=3}^{k} t_m\, \sigma_m(B) \ge -\frac{1}{4} t_2\, Q, \tag{14}$$

the two halves of $1/2 t_2 Q$ splitting between the claimed bound and the allowance. It is (14) that we prove.

The core expansions. On the slice each $\sigma_m(B)$ collapses onto a small set of invariants: expanding by Möbius inversion over ordered pairs of set partitions of $[m]$, a partition with a singleton block contributes a row or column sum of

B and dies, so only those orbit invariants survive whose bipartite multigraph has minimum degree at least two. This expansion, and the reductions at degrees two, three and four, are due to McCullagh [19, §3]; the degree-five reduction specialises his general formula. For B with vanishing line sums,

$$\sigma_2(B) = \frac{1}{2}Q, \quad \sigma_3(B) = \frac{2}{3}p_3, \quad \sigma_4(B) = \frac{3}{2}p_4 + \frac{1}{8}Q^2 + \frac{1}{4}Z - \frac{3}{4}Y_R - \frac{3}{4}Y_C, \tag{15}$$

$$\sigma_5(B) = \frac{24}{5}p_5 + \frac{1}{3}Q\,p_3 + \Gamma_a + 2\Gamma_b - 4\Gamma_c - 4\Gamma_c', \tag{16}$$

the coefficient of p_m being $(m-1)!/m$, the only contributing partition pair being the maximal one in each coordinate. Kernel-checked: the σ_4 expansion in (15) holds at *every* centred B — `sigma_four_centred` in `SigmaFour.lean` (commit `365e44e`), stated with both marginal hypotheses and consuming both, so it holds wherever the row and column sums vanish and not only on $\Omega_n - J_n/n$.

> **Lemma S2 (four a-priori facts).** Let $A \in \Omega_n$, $B = A - J_n/n$. Then *(F1)* $-1/n \le b_{ij} \le 1 - 1/n$; *(F2)* $M \le 1 - 1/n$; *(F3)* $Q \le n-1$; *(F4)* $\|B\|_{\mathrm{op}} \le 1$, and consequently $Z \le Q$.

Proof. (F1) Each entry of A lies in $[0,1]$, being non-negative and at most its row sum; subtract $1/n$. (F2) $q_i = \sum_j A_{ij}^2 - 1/n$ and $\sum_j A_{ij}^2 \le \left(\max_j A_{ij}\right)\sum_j A_{ij} \le 1$ by (F1); columns likewise. (F3) Sum (F2) over i; equality at every permutation matrix. (F4) With $u = n^{-1/2}\mathbf{1}$ we have $Bu = 0$, and for $x \perp u$, $(J_n/n)x = 0$ so $Bx = Ax$; by Birkhoff's theorem A is a convex combination of permutation matrices, each an isometry, so $\|Ax\| \le \|x\|$, and decomposing an arbitrary x gives $\|Bx\| \le \|x\|$. Then $Z \le \left\|B^T\right\|_{\mathrm{op}}^2 \|B\|_F^2 \le Q$. ■

All four are attained at the permutation matrices, so none can be improved by a constant factor. Birkhoff's theorem is where double stochasticity is consumed, and (F4) is the fact that fails first off the slice.

The estimates below are one-sided by design: each invariant is bounded only on the side the sign of its coefficient requires, and the asymmetry of the range in (F1) makes the required side the cheap one. This is where the entry bound earns its keep — for $b \in [-1/n, 1-1/n]$,

$$b^3 \ge -\frac{1}{n}b^2 \quad \text{and} \quad b^3 \le \left(1 - \frac{1}{n}\right)b^2, \tag{17}$$

the first because $b^3 \ge 0$ when $b \ge 0$ and $b^3 = b \cdot b^2 \ge -b^2/n$ when $b < 0$, the second symmetrically. Summing the first gives $p_3 \ge -Q/n$, a factor n stronger than the two-sided $|p_3| \le \beta Q$, which would be too weak for what follows.

> **Lemma S3 (per-invariant estimates).** On the centred slice: *(a)* $p_3 \ge -Q/n$; *(b)* $p_5 \ge -Q/n^3$; *(c)* $p_4 \le \beta^2 Q$; *(d)* $Y_R \le MQ$ and $Y_C \le MQ$ *(attained)*; *(e)* $\Gamma_c, \Gamma_c' \le \beta MQ$; *(f)* $|\Gamma_a| \le MQ$; *(g)* $|\Gamma_b| \le \beta Q$; *(h)* $Qp_3 \ge -\beta Q$; *(i)* $Z \le Q$ *(attained)*; *(j)* $p_4, Q^2, Z \ge 0$.

Proof. (a) is (17) summed, and (b), (c) are the same argument at the other powers, using $b^3 \ge -n^{-3}$ and $b^2 \le \beta^2$ from (F1). (d) $Y_R \le (\max_i q_i)\sum_i q_i \le MQ$, with equality when all q_i agree, in particular at every permutation matrix. (e) By (17), $f_3(i) \le \beta q_i$ with $q_i \ge 0$, so $\Gamma_c \le \beta Y_R \le \beta MQ$. (f) $|q^T B q'| \le \|q\|\|B\|_{\mathrm{op}}\|q'\| \le MQ$ by (F4) and (d). (g) By Cauchy–Schwarz and (F4), $|\Gamma_b| \le \sqrt{p_4}\|B\|_{\mathrm{op}}\left\|B^T B\right\|_F \le \sqrt{p_4}\sqrt{Z} \le \beta Q$ using (c) and (i). (h) Multiply (a) by $Q \ge 0$ and apply (F3). (i) is (F4); (j) is clear. ■

Proof of Theorem G. Combining the core expansions with Lemma S3 term by term — discarding the non-negative invariants that carry a plus sign, by (j) — gives $\sigma_m(B) \ge -C_m(n)Q$ with

$$C_3(n) = \frac{2}{3n}, \quad C_4(n) = \frac{3}{2}\beta, \quad C_5(n) = \frac{24}{5n^3} + \frac{10}{3}\beta + 8\beta^2. \tag{18}$$

For C_4 the only negative terms of (15) are $-\frac{3}{4}(Y_R + Y_C)$, bounded by $\frac{3}{2}MQ \le \frac{3}{2}\beta Q$ by (d) and (F2); for C_5 the six terms of (16) contribute $24/(5n^3)$, $\beta/3$, $M \le \beta$, 2β, $4\beta M \le 4\beta^2$ and $4\beta M \le 4\beta^2$ by (b), (h), (f), (g) and (e). By (14) it therefore suffices that $\Phi(n,k) := (4/t_2)\sum_{m=3}^{k} t_m C_m(n) < 1$, and by the ratio law (12) this is an explicit rational function of n for each k:

$$\Phi(n,3) = \frac{8}{3(n-2)^2}, \quad \Phi(n,4) = \frac{16}{3(n-2)^2} + \frac{12n(n-1)}{(n-2)^2(n-3)^2}, \tag{19}$$

and $\Phi(\cdot,5)$ likewise, its third term $24n^3C_5(n)/(n-2)^2(n-3)^2(n-4)^2$. At $k=2$ the sum is empty, so $\Phi(n,2) = 0$ and the conclusion holds for every $n \ge 2$. Writing each $\Phi(\cdot,k)$ in lowest terms as a ratio of integer polynomials, the denominators $3(n-2)^2$, $3(n-2)^2(n-3)^2$ and $5(n-2)^2(n-3)^2(n-4)^2$ are positive for $n > 4$, so $\Phi(n,k) < 1$ is equivalent to $P_k(n) > 0$ with P_k the denominator minus the numerator:

$$P_3 = 3n^2 - 12n + 4, \quad P_4 = 3n^4 - 30n^3 + 59n^2 - 48n - 36, \tag{20}$$

$$P_5 = 5n^6 - 90n^5 + 445n^4 - 1760n^3 + 620n^2 + 2400n - 3456. \tag{21}$$

Substituting $n = m + n_k$ with n_k the threshold makes every coefficient non-negative with the constant term positive — $P_3(m+4) = 3m^2 + 12m + 4$, $P_4(m+8) = 3m^4 + 66m^3 + 491m^2 + 1280m + 284$, and $P_5(m+14) = 5m^6 +$

$330m^5 + 8845m^4 + 121160m^3 + 861620m^2 + 2716720m + 1660864$ — which settles every $n \ge n_k$ at once, with no appeal to numerics. The thresholds are sharp *for this argument*: $P_3(3) = -5$, $P_4(7) = -568$ and $P_5(13) = -306876$. ■

The binding term throughout is Y_R (with Y_C), whose estimate (d) is attained at the permutation matrices, so the thresholds cannot be lowered by sharpening (d) — only by retaining the three non-negative terms of (15) that the proof discards, and that route is bounded. *Proposition S6*, verified with its witness stored, is that σ_4 on the slice is *indefinite*: an explicit rank-two matrix at $n = 4$ gives $\sigma_4(B)/Q = -1/32$. So $\sigma_4 \ge 0$ is false, and the route can at best replace C_4 by some $\varepsilon \ge 1/32$, worth three steps at $k = 4$ and two at $k = 5$. Proving any such ε is a separate problem, parked here.

> **Proposition S4.** For $n = k = 3$ the conclusion of Theorem G is false.
> Kernel-checked: `not_stabilityAt_three_three` in `TverbergStability.lean` (commit `1507013`), with the witness stored as `witness33` rather than cited.

Proof. Take $A = \frac{1}{2}(J_3 - P)$ for a permutation matrix P of order 3, the doubly stochastic matrix uniform off a permutation. Then $\mathrm{per}(J_3 - P) = 2$, so $\sigma_3(A) = \mathrm{per}(A) = 1/4$, while $\binom{3}{3}^2 3!/3^3 = 2/9$; hence $F = 1/36$. Also $Q = 1/2$, so $F/Q = 1/18$, whereas $c(3,3) = t_2/4 = 1/12 > 1/18$. ■

This is the only excluded cell known to be a counterexample. For $(k = 4, 4 \le n \le 7)$ and $(k = 5, 5 \le n \le 13)$ the inequality is not decided here, and no claim is made either way.

> **Proposition S5 (sharpness).** Let $k \ge 3$ and let $c_{\mathrm{opt}}(n,k)$ be the largest constant for which the conclusion of Theorem G holds. Then in the range covered by Theorem G, $\frac{1}{4}t_2 \le c_{\mathrm{opt}}(n,k) < \frac{1}{2}t_2$.

Proof. The lower bound is Theorem G. For the upper bound take $B = J_n/n - P$ and $A_s = J_n/n + sB$, doubly stochastic for $0 \le s \le 1/(n-1)$. The entries of B are $-\beta$ at the n cells of P and $1/n$ elsewhere, so $p_3(B) = ((n-1)/n^2)(1 - (n-1)^2) < 0$ for $n \ge 3$, and by Lemma S1 and the core expansions $F(sB)/\|sB\|_F^2 = \frac{1}{2}t_2 + st_3\big(\frac{2}{3}p_3(B)\big)/Q_B + O(s^2)$, strictly less than $\frac{1}{2}t_2$ for all small $s > 0$. ■

So the constant of Theorem G is optimal to within a factor two, and that factor cannot be closed: no constant as large as $\frac{1}{2}t_2$ works for any $k \ge 3$. A verified computation, not a theorem, corroborates the picture at $k = 4$: the largest admissible constant, measured against the claimed $c(n,4)$, is 1.88, 1.91, 1.93 at $n = 8, 9, 10$ — rising toward the ceiling of Proposition S5 and never reaching it.

Verification. Every displayed identity and estimate of this subsection is checked over $\mathbb{Q}$, with no floating-point arithmetic in any decision, by the standalone script `graded_verify_stability.py`, which imports nothing beyond the standard library: Lemma S1 and the core expansions against brute-force subpermanent sums; the four facts of Lemma S2; each estimate of Lemma S3 in the one direction the proof uses, reporting its slack ratio so the two tightness claims are checkable; the per-layer bounds (18); the closed forms for Φ, the thresholds and the exceptional sets; the polynomial argument term by term, including the sign pattern of $P_k(m + n_k)$ and the sharpness values $P_k(n_k - 1) \le 0$; Theorem G end to end at every covered cell; Propositions S4 and S5; and the formalisation scope of §5 against the Lean sources at the commits cited there, read with `git show` so the working tree cannot flatter the claim. That last check is the one exception to the script being self-contained, and run with `--no-lean` it skips it and *records in its log that the scope claims went unchecked.* The test matrices are the configurations at which Lemma S3 is attained or nearly attained. Four mutation controls fire — the constant doubled, a sign flip in (15), each threshold lowered by one so that an excluded cell is claimed as covered, and Lemma S3(a) over-tightened — and with no fault injected those same four checks raise nothing, so each rejection is attributable to its fault.

3.3 The collar assembly at $k = 4$

Let $A \in K_n$ and $B = A - J_n/n$ with row sums R and column sums C. Set $x_i = R_i/n$, $y_j = C_j/n$, $L_{ij} = x_i + y_j$ and $z = B - L$, so $\sum_i x_i = \sum_j y_j = 0$, z is doubly centred, and $\|L\|_F^2 = n(\mu + \nu)$ with $\mu = \|x\|^2$, $\nu = \|y\|^2$. The invariants Q, q_i, q_j', f_3, g_3 are as in §3.2 but now taken of z, and $N = zz^T$, $N' = z^T z$. Confinement bounds the line block only: $\mu + \nu \le u_{\max}(n,k) = (n-1)k!/n^{k+1}$.

What couples. (3) gives $F = \sum_d [t_d \sigma_d(B) - s_d(e_d(R) + e_d(C))]$, whose e_d half never couples, since z has vanishing line sums and so the line sums of $L + z$ are those of L alone. For the σ_d half, expansion by the number of z factors,

$$\sigma_d(X + Y) = \sum_{j=0}^{d} \sum_{|S| = |T| = j} \mathrm{per}(X[S|T])\, \sigma_{d-j}\big(Y^{(S,T)}\big), \tag{22}$$

with $Y^{(S,T)}$ being Y with rows S and columns T deleted, taken at $X = z$, $Y = L$, gives $F(L + z) = F_{\mathrm{line}}(x,y) + F_{\mathrm{centred}}(z) + \sum_d t_d \cdot$ (cross parts of σ_d). At $d = 2$ the cross part vanishes *identically*, not merely to leading order.

The five cross terms. All are exact, and all are verified against brute force. At $d = 3$,

$$X_1 = (n-2)^2\, x^T z\, y, \quad X_2 = (2 - n)\Xi, \quad \Xi := \sum_i x_i q_i + \sum_j y_j q_j'; \tag{23}$$

at $d=4$,

$$Y_1 = -(n-2)(n-3)^2\big[\,x^T z(y\circ y) + (x\circ x)^T z\,y\,\big], \tag{24}$$

$$Y_2 = (n-2)(n-3)\left[-\sum_{s_1<s_2} N_{s_1s_2}e_2(x_I) - \sum_{t_1<t_2} N'_{t_1t_2}e_2(y_J)\right] + 2(n-3)^2\sum_{i,j} z_{ij}^2 x_i y_j, \tag{25}$$

$$Y_3 = -(n-3)[2A_1 - A_2 + 2A_3 - A_4],\quad A_1 = \sum_a x_a f_3(a),\quad A_2 = x^T z\, q', \tag{26}$$

$A_3 = \sum_b y_b g_3(b)$, $A_4 = q^T z\, y$, where $\circ$ is the entrywise product and $I=[n]\setminus S$, $J=[n]\setminus T$ for the deleted index pairs. One principle produces all three of the $d=4$ terms: *anything separable in the deleted indices is annihilated by* $\sum_a z_{ab} = \sum_b z_{ab} = 0$. For Y_1 what survives is that the restricted e_1 is *not* zero — on $I=[n]\setminus\{a\}$ one has $e_1(x_I) = -x_a$ and $e_2(y_J) = e_2(y) + y_b^2$ — so the terms depending on one deleted index alone die and the rest leave $x_a y_b^2$ and $x_a^2 y_b$. For Y_2 it is a vanishing lift: the e_2 pieces depend on S alone and T alone, with $\sum_T \operatorname{per}(z[S|T]) = -N_{s_1s_2}$, while for the $x_S y_T$ piece the lift of the restricted sum to the unrestricted one is *identically zero*, every expanded piece carrying a lone $\sum_s z_{st}$ or $\sum_t z_{st}$, so the restricted sum is a pure diagonal correction — and an absolute-value bound, discarding exactly that cancellation, overshoots by four orders of magnitude. For Y_3 it is subpermanents through a row: $\sigma_1\big(L^{(S,T)}\big) = -(n-3)(x_S + y_T)$ and $\sum_{S,T}\operatorname{per}(z[S|T])x_S = \sum_a x_a\big(2f_3(a) - (zq')_a\big)$.

The merge, and why $k=4$ inherits the clean form. On the collar $A\ge 0$ gives the *per-entry* bound $z_{ij} \ge -\big(1/n + x_i + y_j\big)$, not $z_{ij}\ge -1/n$; cubing and summing gives $\sum_{ij} z_{ij}^3 \ge -Q/n - \Xi$, whose perturbation is *exactly* the invariant of the cross term X_2. The two effects are therefore one quantity, to be charged once — but *at its full summed coefficient*: the $m=3$ layer contributes $\frac{2}{3}\Xi$ and the cross term $(n-2)\Xi$, so the coefficient is $(3n-4)/3$, not $(n-2)$. Counting once deletes the double count, not a coefficient. At $k=4$ the layers are $m=2,3,4$ and the only odd-power one-sided step is at $m=3$: the centred core of σ_4 is (15), every term of even degree, so no second perturbation can arise and the merge keeps its single-coefficient form. This is special to $k\le 4$: at $m=5$ the perturbation enters as $\big(1/n + x_i + y_j\big)^3$, which is four cross invariants rather than one.

The collar facts. Write $\rho^2 = (n-1)k!/n^{k-1}$ for the confinement radius on line sums and $\beta_c = 1+\rho-1/n$. The four facts below are the collar forms of (F1)–(F4), in the same order, with the degradation the collar costs. *(K1)* $-1/n \le b_{ij} \le \beta_c$; the lower side needs only $A\ge 0$ and so does *not* degrade off the slice, and that asymmetry is what makes the $m=3$ step cheap. *(K2)* $\max_i q_i(B) \le (1+\rho)^2 - 1/n + 2\rho/n$. *(K3)* $Q \le n-1+\rho^2$. *(K4)* $\|B\|_{\mathrm{op}} \le \sqrt{(1+\rho)^2 + \rho^2/n}$, since a non-negative matrix with all line sums at most $1+\rho$ has operator norm at most $1+\rho$: Birkhoff is lost off the slice, but the bound is not. Consequently $\|z\|_{\mathrm{op}} \le \|B\|_{\mathrm{op}} + \sqrt{2n\,u_{\max}}$.

The one unsettled constant. The slice bound $q_i(z)\le 1-1/n$ requires row sums equal to 1 *and* non-negativity; on the collar $J_n/n + z$ has row sums 1 but may have negative entries, so the bound does not transfer, and it is refuted — a permutation with one reweighted row violates it by a factor 1.63 at $(k=4, n=10)$. We therefore write $q_i(z) \le c\,(1-1/n)$ and carry c symbolically. A verified sensitivity computation, graded as exactly that and recorded with its ten sampled values in the reproduction kit, recomputes the honest threshold exactly over $\mathbb{Q}$ at each and finds it equal to 10 at every sampled value inside the admissible band $1.58 \le c \le 2.53$, whose ends are the smallest constructed violation and the proved collar cap; no claim is made for unsampled c.

The budget. Layer two is the budget: on the centred block $t_2\sigma_2(z) = \frac{1}{2}t_2 Q$, and on the line block $F_{\mathrm{line}} \ge \frac{1}{2}\lambda_{\mathrm{line}}(\mu+\nu)$. Every other contribution is bounded below and charged to whichever budget it scales with — X_1 and Y_1, Y_2 carry $(\mu+\nu)$ or $(\mu+\nu)^{3/2}$ and go to the line side; Ξ and Y_3 carry Q or $\sqrt{\mu+\nu}\sqrt{MQ}$ and go to the centred side, Y_3 split between the two by the arithmetic–geometric mean. Both totals below 1 is the conclusion, and that is Theorem H.

Verification. `graded_verify_k4.py` recomputes every displayed quantity of this subsection over $\mathbb{Q}$ with no floating-point arithmetic in any decision: (22) at $d=2,3,4$ against brute force, including that the $d=2$ cross part is exactly zero; the end-to-end split identity with every t_d in place, against F computed from the 1992 functional; all five cross-term reductions against brute force; the per-entry bound and the merge, on configurations of *both* signs of $\sum z^3$; the four collar facts; every budget line recomputed from the layer identity; and the sensitivity table row by row, parsed from its committed source rather than restated, so displayed and checked cannot drift apart. Four mutation controls fire, each with a *separating* witness asserted in the same line: a t_d factor dropped from a cross term, caught by the end-to-end identity; the merge coefficient $(3n-4)/3$ replaced by $(n-2)$, caught by the budget-line check; the $d=2$ cross part assumed non-zero, caught by the expansion check; and c rescaled in a c-independent line, caught by the sensitivity audit. Two of the four encode errors made and corrected while the argument was being assembled — a dropped t_4, and the merge coefficient. They are controls because they happened.

4 The small cells at $k=4$, and the gap

Theorem H begins at $n=10$. The five cells below it are each a fixed-n question, recorded at the grade each has actually reached; no cell inherits the grade of a neighbour. *Anchor grade* means an exact rational certificate accepted in full by six checks of the standalone verifier — the bound; σ_k by two structurally different algorithms; the identity by *full* coefficient comparison; definiteness of the assembled Grams by exact LDL^T over $\mathbb{Q}$; equality only at J_n/n; mutation tests rejected — and anything less is stated as what it is.

cell	status	support, and what is missing
$(k=4, n=5)$	settled: $\Phi_4 \leq 1226/625$ on K_5	anchor grade; 26 exact LDL^T factorisations of size 350, two independent full runs logged
$(k=4, n=6)$	settled: $\Phi_4 \leq 107/54$ on K_6	anchor grade on a stored exact witness: identity over all 40,386 coefficients, both assembled Grams (cone size 702) definite over $\mathbb{Q}$, conjugacy proved exactly
$(k=4, n=7)$	settled: $\Phi_4 \leq 4778/2401$ on K_7	anchor grade on a stored exact witness: identity over all 156,555 coefficients, conjugacy exact, assembled positivity by the congruence route below
$(k=4, n=8)$	*not settled*; one check short	five of the six checks hold on a stored exact witness — identity over all 496,448 coefficients, the bound 1021/512, equality only at $J_8/8$, conjugacy exact, and positivity *block-wise* for all 21 canonical blocks. The assembled 2144×2144 factorisation has not run, and block definiteness is not assembled definiteness
$(k=4, n=9)$	*not settled*	no witness exists: the exact solve exceeded available memory at this size. Nothing is stored, and nothing is claimed

Table 3: The five cells between Theorem A's line and Theorem H's range, as of 30 July 2026. In every settled cell equality holds only at J_n/n.

At $(k=4, n=7)$ both assembled Grams are positive definite over $\mathbb{Q}$ by *congruence* rather than by a single direct factorisation: the 21 isotypic components are exactly H-orthogonal over $\mathbb{Q}$ and their translates span; on each component the congruence block is a Kronecker product $C_b \otimes h_b$, and all 21 of each factor are positive definite under exact LDL^T; the Kronecker argument then makes each component block definite, orthogonality-with-spanning assembles the components, and the conjugacy extends the verdict to all 49 multiplier Grams. Every link is a stored artefact, verified exactly over $\mathbb{Q}$ with rejection controls that fire. The two component computations use different bases of the same components, and the check reconciling them is a *consistency check between stored artefacts, not an independent verification.*

The gap, stated once. Unconditionally, the line $k=4$ is settled for $n=5,6,7$ at anchor grade and for every $n \geq 10$ by Theorem H, and is *not settled at* $n=8$ *and* $n=9$. It becomes settled for every $n \geq 5$ exactly when those two cells reach at least anchor grade, and not before. No theorem of this paper depends on any cell of Table 3.

Two routes could close it: fixed-n certificates at the two open cells, or a certificate family in $\mathbb{Q}(n)$ at $k=4$, which would need no anchors and would besides supersede the local route and inherit the $k=3$ formalisation pattern. The groundwork for the second is a decided sweep rather than a certificate, recorded in the kit with its witnesses; two of its facts bear on the method's reach. Mixed Gram degrees are ruled out for even k, since they would force the top form F_k to be a sum of squares on the hyperplane, and (3) gives $F_k(b) = s_k[s_k\sigma_k(b) - e_k(R) - e_k(C)]$, negative at explicit doubly centred integer matrices; so the programme's counts are constant on the bands $\{2,3\}$, $\{4,5\}$, $\{6,7\}$, and grow from 19 unknowns at $e=1$ to 440 at $e=2$, which is why the route reaches every *fixed* k at all large n and cannot reach $k=n$. And for band two the pinning programme is decided over $\mathbb{Q}$ at $n=5,6,7$, with no verdict resting on a solver's reported sign. There is still no certificate family in n for band two, so there is nothing yet to formalise.

5 What is machine-checked, and what is not

> The development elaborates with no `sorry`, and no declaration in it depends on `sorryAx`. `native_decide` is used nowhere. Every declaration cited above in support of a stated result carries a `#print axioms` command, and each returns exactly `[propext, Classical.choice, Quot.sound]` — with one exception, `NewtonIneq.choose_mul_sub`, a statement about binomial coefficients, which returns the strict subset `[propext, Quot.sound]`. Build success is not an axiom audit, and the two are kept separate here.

The $k=3$ chain. On the Lean sources as committed at `32c811e`, the eight files carrying Theorems A and C–F carry `#print axioms` on 511 declarations: `SubDittertK3.lean` and `RookSum.lean` (Theorem A), `SubDittertUniversal.lean` (C), `SubDittertK2.lean` (D), `SubDittertLinear.lean`, `SubDittertM.lean` and `SubDittertMaclaurin.lean` (E), and `NewtonInequalities.lean` (F), built against Lean 4.14.0 and Mathlib `v4.14.0` [16]. In `SubDittertK3.lean` the 377 are *every* named declaration in the file, private ones included. Elsewhere the audits cover the results and their supporting lemmas rather than every private definition, which is not a gap: `#print axioms` reports the axioms of the *transitive closure* of a proof term, so auditing a theorem audits everything it rests on, and no file contains a `sorry`, so nothing can acquire `sorryAx` from within.

`subDittert_k3_full` is Theorem A in all three parts, with `Kn`, `esym`, `subPerm`, `sigmaK`, `E`, `P` and `Phi` built from scratch inside Lean and the final statement written in the 1992 notation. Two declarations *validate* those definitions, and they matter more than they look: `Phi_uniform` proves $\Phi_k(J_n/n) = 2 - k!/n^k$ for every $k \leq n$, so the functional as formalised is tight at the conjectured maximiser exactly where the conjecture says it is; and `sigmaK_rankOne` proves $\sigma_k(xy^T) = k!e_k(x)e_k(y)$, the check that rows are read from S and columns from T — a definition reading both indices off the same subset satisfies a different identity, and a symmetric test matrix cannot detect that error. From there, `RookSum.lean` and `obj_eq_objPoly` close the combinatorial half for all n at once, eliminating `Matrix.permanent`, `sigmaK`, `esym` and every `Finset.powersetCard` sum from everything downstream; `certPositive_of_four_le` proves the ten positivity facts for every *real* $n \geq 4$; `G0_posSemidef` and `Hm_posSemidef` give both Gram families with no spectral theorem, and `G0_posDef` the definiteness the equality case needs; `certificate_identity` is the Positivstellensatz identity itself; and `subDittert_k3_stability` with `eq_uniform_of_Phi_eq_of_stability` give the stability bound and the equality case, which stands on two independent Lean proofs.

Theorem G's cells. These rest on further files, counted at their own commits and held to the same standard: `StabilityK3.lean` (33 audit lines) and `TverbergStability.lean` (21) at `1507013`, `LayerIdentity.lean` (28) at `27bda3f`, `SigmaFour.lean` (46) at `365e44e`, `StabilityK4.lean` (14) at `944b517`, and the partial `StabilityK5Atoms.lean` (15) at `f2bdc7f`. Theorem G is kernel-checked at $k = 2$ for every $n \geq 2$, at $k = 3$ for every $n \geq 4$, and at $k = 4$ for every $n \geq 8$ — in each case the whole range the statement covers for that k, with no arithmetic gap between the formalised and the written range. All three discharge `TverbergStability.StabilityAt k n`, the displayed inequality of Theorem G with `cVal` equal to $c(n, k)$; the $k = 4$ proof follows this paper's route step for step, its threshold arithmetic consumed as `Phi4_lt_one` with the bridge `layer_ratio_lt` checking that Φ rebuilt from the Lean coefficients is exactly (19). Lemma S1 is kernel-checked at *every* k, under hypotheses weaker than this paper's — only $1 \leq k \leq n$ and the single scalar constraint $\sum_{ij} B_{ij} = 0$, where Lemma S1 as stated here assumes A doubly stochastic — so the identity underneath the whole argument holds formally on a strictly larger set than the theorem needs. Proposition S4 is kernel-checked with its witness stored rather than cited, as is the threshold layer at $k = 3, 4, 5$; and `three_threshold_not_slack` gives the sharpness of the $k = 3$ threshold in the strong form that assuming stability at every $n \geq 3$ is contradictory.

Two limits of the $k = 4$ formalisation must not be over-read. First, the threshold 8 is *a limit of the argument, not known sharp*: the cells $(k = 4, 4 \leq n \leq 7)$ are undecided, and there is no $k = 4$ analogue of `three_threshold_not_slack` — nothing here should be read as implying one. Second, the binding atom is Y_R (with Y_C), attained at the permutation matrices, so the threshold cannot move by sharpening anything in the formalised file.

What is not machine-checked. Theorem H, and with it everything the collar costs on top of the slice — the collar facts, the cross-term reductions, the merge and the budget — rests on written proofs plus the exact verifier. At $k = 5$ neither the estimates feeding the budget nor Theorem G itself is formalised: there is no `stabilityAt_five`, and while five of the $k = 5$ per-invariant atoms *are* kernel-checked, the degree-five expansion, estimates (f) and (g), fact (F4) and the assembly are not, so the cell is not closed — partial atom coverage is not cell coverage, and we do not average the two. The fixed-dimension certificates of §4 are unformalised, as is the block-diagonalisation of §2.3 *as an equivalence*: Lean proves the implication the theorem needs, by explicit algebra rather than by that route, so the reduction is corroborated by the Lean development without being verified by it. The constant $\theta_2(n)$ is not claimed optimal — measured slack ratios of 2.88, 4.34, 5.70 at $n = 4, 5, 6$ put it within a factor of three to six of the best constant where it is closest to tight. Finally, what the kernel cannot check anywhere is that the Lean definitions say what the 1992 paper says; the validation step above is the argument that they do, and it is the only place left where a human error would survive the machine.

How the division is checked, not asserted. The stability verifier reads each Lean file at the commit cited for it and confirms both halves: that `cVal` *is* $c(n, k)$, compared as exact rationals over a grid rather than as text; that the hypotheses of `stabilityAt_two`, `stabilityAt_three` and `stabilityAt_four` are the thresholds claimed above; that `sigma_four_centred` carries both marginal hypotheses; that Lean's P_3, P_4, P_5, an independent derivation, agree at every point with the polynomials the verifier builds; and, on the negative side, that no `stabilityAt_five` exists — with a non-vacuity check that the same search does find the three theorems that are there, since an absence found by a broken search is not an absence. It also runs the orphan diff on the four files carrying audit blocks, empty in both directions: 33 of 33 declarations in `StabilityK3.lean` carry `#print axioms` lines, as do 14 of 14 declarations in `StabilityK4.lean`, 46 of 46 declarations in `SigmaFour.lean` and 28 of 28 declarations in `LayerIdentity.lean`, and no source contains `sorry` or `native_decide` outside its comments. All four counts are parsed back out of this section, so a number stated here and a number measured there cannot drift apart. One limit is worth stating: the axiom sets themselves — that each of the 121 audited declarations reports a subset of `propext`, `Classical.choice`, `Quot.sound` with no `sorryAx` — come from elaborating the files, which only Lean can do and which the verifier does not repeat.

What is not claimed, and where the methods stop. At $k = 4$: nothing beyond $n \geq 10$ and the settled cells $n = 5, 6, 7$. At $k = 5$: nothing beyond the stability cell $(k = 5, n \geq 14)$ at written-proof plus exact-verifier grade — the Cheon–Hwang inequality itself is not claimed at any $(k = 5, n)$. No statement about general k beyond Theorems C and E, and no novelty for the identity of Theorem C, the statement of Theorem D, or Theorems E and F. Nothing refereed. No floating-point number enters any decision above: the numerics find candidates, and every accepted statement is re-established over $\mathbb{Q}$ or $\mathbb{F}_p$. As for reach, $\deg F = k$, so the Gram basis degree must grow with k and the degree-counting obstruction that blocks the $k = n$ line reappears in k; the line $k = 3$ is tractable exactly because its degree budget is fixed while the dimension grows. The local route stops where its merge does — at $m = 5$ the one-sided step splinters into four cross invariants, so $k = 5$ needs new ideas on the centred block rather than more of the same. Two problems are parked with their obstructions priced: a lower bound $\sigma_4 \geq -\varepsilon\|z\|_F^2$ on the doubly centred slice for some $\varepsilon \leq 1/16$, which would carry Theorem H down to $(k = 4, n \geq 8)$ and is known false at $\varepsilon = 0$; and a certificate family in $\mathbb{Q}(n)$ at $k = 4$. To our knowledge Theorem A is the first resolved case of the Cheon–Hwang conjecture with $2 < k < n$ and Theorem G the first stability form of the Tverberg–Friedland theorem; both claims cover public code repositories as well as the indexed literature, both are priority claims rather than claims of difficulty, and both are perishable, so they should be re-checked before this paper is submitted anywhere.

6 Data availability

The certificates as exact rational data, the objective builders, the block-diagonalisation, the Sturm decision, the independent verifiers with their controls, the sensitivity record of §3.3, the stored exact witnesses behind Table 3 with the artefacts of the congruence route, the band-two sweep, and the Lean development are supplied as a reproduction kit, whose README names the claim each file backs and the procedure for re-running each check. The failed design branches are retained there, because the four closed routes of §2.3 are only checkable against them.

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